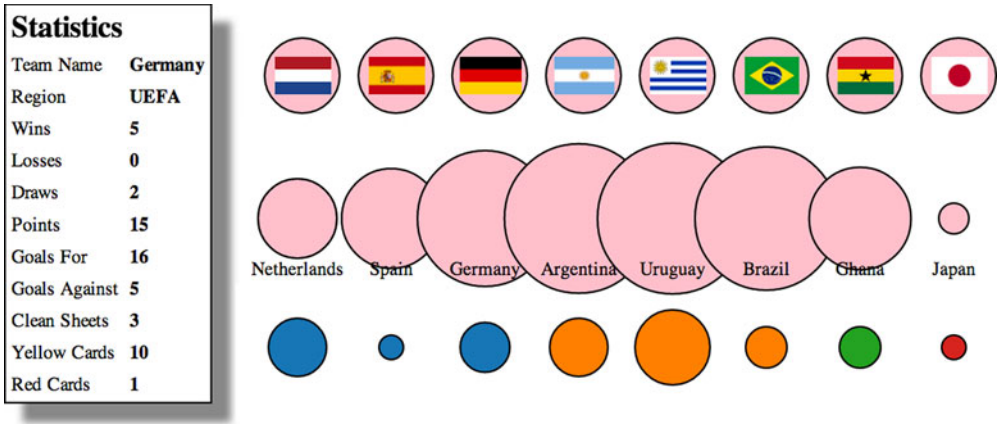


Figure 3.1 This chapter covers loading HTML from an external file and updating it (section 3.3.2), as well as loading external images for icons (section 3.3.1), animating transitions (section 3.2.2), and working with color (section 3.2.4).

This chapter deals with design broadly speaking, and it touches not only on graphical design but on interaction design, project architecture, and the integration of pregenerated content. It highlights the connections between D3 and other methods of development, whether we're identifying libraries typically used alongside D3 or integrating HTML and SVG resources created using other tools. We can't cover all the principles of design (which isn't one field but many). Instead, we'll focus on how to use particular D3 functionality to follow the best practices established by design professionals to create some simple data visualization based on the statistics associated with the 2010 World Cup, as seen in figure 3.1.

3.1 Project architecture

When you create a single web page with an interesting information visualization on it, you don't need to think too much about where all your files are going to live. But if you build an application that provides multiple points of interaction and different states, then you should identify the resources that you need and plan your project accordingly.

3.1.1 Data

Your data will tend to come in one of two forms: either dynamically delivered via server/API or in static files. If you're pulling data dynamically from a server or API, it's possible that you'll have static files as well. A good example of this is building maps, where the base data layer (such as a map of countries) is from a static file and the dynamic data layer (such as the places where tweets are made) comes from a server. For this chapter, we'll use the file worldcup.csv to represent statistics for the 2010 World Cup: